cahier des charges – Kado

Digital discount coupons from local shops, pharmacies, and restaurants — save money on everyday essentials and corporate dining, even without insurance or internet.

0. Document Header

- **Project Title:** **Kado** — *A local coupon web app where businesses (shops, pharmacies, restaurants) create digital vouchers, and anyone can discover, save, and redeem them offline.*
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- **Date:** 17 June 2026
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1. Presentation and Context

In Cameroon, households and even company employees often struggle to afford quality meals, medicines, and daily goods. Restaurants lose potential corporate group bookings because there's no easy way to target nearby offices with a lunch discount. Small shops and pharmacies lack a simple digital tool to reach price-sensitive customers.

Kado bridges this gap. It is a Progressive Web App where any registered business — a bakery, a pharmacy, or a restaurant — can create a digital coupon in seconds. A restaurant can offer "20% off for groups of 10+" to attract corporate teams. A pharmacy can discount essential medicines for those without insurance. Users browse, save, and redeem coupons offline: a QR code or numeric code is generated on the customer's phone and verified by the cashier even without internet.

This project addresses **digital inclusion, local commerce, public health, and corporate dining affordability** — all without requiring bank cards, insurance policies, or high-end smartphones.

2. Problem and Objectives

Problem:
Cost-conscious families skip meals out and needed medication because they can't find verified discounts. Restaurants can't easily offer targeted lunch deals to nearby companies. Small businesses have no digital tool to publish and track promotions, forcing them to rely on fleeting WhatsApp posts.

Main Objective:
Provide a lightweight web app where any local business — especially restaurants, pharmacies, and shops — can publish a digital coupon, and any customer can save and redeem it offline, in under 3 taps.

Secondary Objectives (measurable):

- Enable a restaurant to create a “corporate lunch” coupon in under 1 minute.
- Guarantee coupon redemption works offline (QR scan or code entry) and syncs when back online.
- Prevent coupon reuse with a central burn mechanism.
- Let users filter coupons by category (pharmacy, restaurant, grocery) and location (optional).

3. Project Scope

Included (In this project)	Excluded (Out of scope)
Business registration (name, type including restaurant, pharmacy, shop, location, phone)	Full e-commerce checkout or food delivery
Coupon creation (discount %, fixed amount, validity, max uses, group size if needed)	Payment processing (coupons are free to use)
User wallet to save and display coupons	Loyalty points or gamification (future)
Offline QR code generation and numeric fallback	Delivery of physical goods or table booking
Merchant verification interface (scan / code entry, offline queue)	Integration with insurance systems or corporate HR portals
Simple moderation flag for fake coupons	Complex analytics dashboard
Responsive design (mobile-first)	Native mobile app builds

4. Target Users (Personas)

Profile	Who they are	What they need
The thrifty parent	Mother of three, market trader, has a smartphone with limited data, no insurance.	Find discounts on food, milk, and basic medicines. Save coupons to show later, even offline.
The office team lead	Works at a bank or telecom in Douala. Organizes monthly team lunches.	Discover restaurants offering “group meal” coupons near the office to save the team money.
The restaurant owner	Runs a mid-range restaurant in Bonapriso or Bastos. Wants to attract corporate groups during weekdays.	Quickly create a coupon like “15% off for tables of 6+” and track redemptions.
The pharmacy owner	Neighbourhood pharmacy. Serves low-income customers.	Create discount coupons on generics and over-the-counter drugs to help

		those without
The small shopkeeper	Owns a grocery stall or bakery. Promotes daily deals.	A digital noticeboard for specials that customers can save and redeem.

5. Prioritized Features (MoSCoW)

Priority	Feature	Why
Must (vital)	1. Business registration & coupon creation (simple form)	No businesses, no coupons
Must	2. Coupon browsing (by category: Restaurant, Pharmacy, Grocery, etc.) and saving to wallet	Discovery and personal collection
Must	3. Offline coupon display with QR code + numeric code	Redemption works without internet
Must	4. Merchant verification interface (scan/code entry, offline queue)	Prevent reuse, enable offline sync
Should (important)	5. Category filters and basic search	Quick discovery
Should	6. Redemption counter &	Prevents abuse

	max-use	
Could (bonus)	**7. Location-based "Near me" filter**	**Corporate users find restaurants nearby**
Could	**8. Push notification for new restaurant/pharmacy coupons**	**Engagement**
Won't (not now)	**Payment integration, table booking, loyalty points, corporate HR integrations**	**Beyond MVP scope**

MVP = only the 4 Must items.

6. User Stories

- As a **restaurant owner**, I want to create a coupon offering 20% off for groups of 8+ people so that I attract corporate lunch bookings.
- As a **pharmacy owner**, I want to create a coupon for 15% off paracetamol so that uninsured customers can afford medicine.
- As an **office team lead**, I want to browse restaurant coupons near my workplace so I can plan an affordable team outing.
- As a **parent**, I want to save a pharmacy coupon to my wallet and show it later (even offline) so I can get a discount when buying medicine.
- As a **cashier**, I want to scan a QR code or type a 6-digit code to mark a coupon as used, even when the internet is down, so the same coupon isn't reused.
- As a **shopkeeper**, I want to register my business and publish a coupon in under 2 minutes so that my promotion reaches digital users.

7. Screen Map and Screens

Main navigation (user side):

Home → Browse Coupons (filter by category: All / Restaurant / Pharmacy / Grocery) → Coupon Detail → Save to Wallet → My Wallet → Coupon Display (QR + code).

Merchant side:

Merchant Login → Coupon Scanner / Code Entry → Verification Result → Redemption History (synced).

Screens:

Screen	Main Content
Home (public)	Search bar, category chips (Restaurant, Pharmacy, Grocery, Bakery, Services), featured coupons.
Coupon List	Cards showing business name, type, discount, validity, “Save” button. E.g., “Chez Hélène – 20% off group lunch (6+ people)”.
Coupon Detail	Full terms, business location on a simple map (static image or coordinates), “Save to Wallet” button.
My Wallet	List of saved coupons with status (active/used/expired). Tap to open.
Coupon Display	Large QR code, 6-digit code, discount details, expiry. Works offline.
Merchant – Scanner	Camera view to scan QR, fallback input for numeric code. “Verify” button.

Merchant – Result	**Success/fail message, coupon details, “Mark as Used” confirmation.**

All screens are responsive, built mobile-first.

8. Data Model

Entity	Fields	Description
Business	id, name, type (restaurant / pharmacy / grocery / bakery / other), phone, address, lat, lng, status (active/flagged)	Registered business
Coupon	id, businessId, title, description, discountType (percent/fixed), discountValue, validFrom, validUntil, maxRedemptions, currentRedemptions, minGroupSize (optional for restaurants), status (active/expired/deleted)	A coupon with optional group size requirement
User	id (device-generated anonymous ID),	Browser-based user

	savedCoupons: [couponId, savedAt]	
Redemption	id, couponId, userId, redeemedAt, code (generated one-time code), status (pending/synced/used)	Each redemption attempt

9. Main User Journey

Scenario: A corporate team lead books a group lunch with a restaurant coupon.

Step	Action
1	Opens Kado, taps “Restaurant” category. Sees “20% off for 8+ people – Chez Hélène”.
2	Taps “Save”. The coupon goes to her Wallet.
3	At the restaurant, she opens her Wallet (no internet needed), shows the QR code.
4	The waiter scans it with the merchant view (works offline). “Valid for group of 8+”.
5	The discount is applied to the bill. The coupon is burned. When the network returns, the redemption syncs to prevent reuse.

10. Technical Specifications

- **Frontend:** Angular v18+, PWA with Service Worker.
- **Styling:** SCSS, Angular Material (or custom components for lightness).
- **Data:** Firebase Firestore for real-time sync; IndexedDB (Dexie.js) for offline wallet and redemption queue.
- **QR generation:** qrcode library (client-side).

- **Authentication:** Anonymous device ID for users; business login via phone number + OTP (optional).
- **Merchant view:** Separate lazy-loaded module or standalone build from the same codebase.
- **Hosting:** Firebase Hosting or Vercel (free tier).

11. Non-functional Requirements

- **Offline resilience:** Wallet and saved coupons work offline. QR codes are generated locally. Redemption requests queue offline and sync later. The app shell and last-browsed coupons are cached.
- **Performance:** First load under 3 seconds on 3G; coupon display renders in under 1 second (cached).
- **Responsive:** Mobile-first; usable from 320px width to desktop.
- **Accessibility:** High contrast, alt text for QR (numeric fallback), simple language.
- **Security:** One-time codes are hashed and validated server-side on sync to prevent forgery. No sensitive data stored.

12. Visual Identity (UI)

Element	Choice
Primary colour	Fresh green #00A86B (savings, health, growth)
Secondary colour	Warm yellow #FFD700 (optimism, visibility)
Typography	Headings: Nunito, Body: Inter
Tone	Friendly, helpful, simple
Logo	A gift box with a heart or a price tag with “%”, in green and yellow. Name “Kado” in bold.

13. Schedule and Milestones

Milestone	Content	Deadline
D1 – Framing	**Cahier des charges + GitHub repo**	**Week 1**
D2 – Mockups	**Wireframes, Angular scaffold**	**Week 1–2**
D3 – MVP	**Must features (business coupons, browsing, wallet, offline QR, merchant scanner)**	**Week 3–4**
D4 – Finishing	**Should features (categories, search, counter), polish, deploy demo**	**Week 5–6**

14. Expected Deliverables

- Source code on GitHub (branches: main, trainer, learner).
- Deployed app URL (Firebase Hosting).
- This specification (README).
- Setup instructions (npm install, ng serve, test service worker).
- Screenshots of key journeys (create a restaurant coupon, save, redeem offline).

15. Success Criteria

- A restaurant can create a “15% off group of 6+” coupon in under 2 minutes.
- A user can filter by “Restaurant”, save a coupon, and view it offline.
- The merchant scanner works offline and syncs redemption after reconnection.
- The same coupon cannot be redeemed beyond its maxRedemptions.
- The app works smoothly on a low-end smartphone on 3G.

16. Risks and Plan B

Risk	Plan B
Firebase quota exceeded during demo	**Fallback to JSON Server or pure IndexedDB mock**
QR scanner fails on some devices	**Fallback to manual 6-digit code entry**
Offline redemption conflict (two cashiers scanning same coupon)	**First-to-sync wins; second gets “Already used” after sync**
Low restaurant sign-up during development	**Pre-seed demo with a few fake restaurant accounts for testing**

17. Appendices

- **Glossary:** *PWA, QR code, IndexedDB, Redemption* (same as before).
- **Inspiration:** Groupon (model), GoodRx (pharmacy discounts), TheFork (restaurant discounts for groups).
- **Wireframe sketches:** to be attached (Figma or hand-drawn).